Literature Review

Neighborhood and Street Quality Effects on Housing Prices

and

Housing Quality as a Component of Effective Housing Supply

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Abstract

This review covers two related bodies of literature. The first examines how neighborhood and street characteristics—transportation access, street connectivity, local amenities, streetscape perception, and walkability—are capitalized into residential property values, primarily through the hedonic price model. The second examines the concept of quality-adjusted or "effective" housing supply within spatial equilibrium models. In labor economics, heterogeneous workers are aggregated into "effective labor" using relative wages as weights. An analogous question arises for housing: can heterogeneous dwelling units be aggregated into "effective housing supply" units using hedonic price weights? We review the theoretical foundations and empirical progress on this question, connecting the micro-level hedonic evidence on neighborhood quality to the macro-level spatial equilibrium literature on housing supply, labor allocation, and equilibrium prices.

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1 Neighborhood and Street Quality Effects on Housing Prices

The relationship between neighborhood characteristics and housing values is one of the most extensively studied topics in urban and real estate economics. Rooted in hedonic price theory—originating with —the literature treats a housing unit as a bundle of attributes whose individual components are capitalized into observed market prices. This section reviews the empirical and theoretical literature on how specific dimensions of neighborhood and street quality affect residential property values.

1.1 The Hedonic Price Framework

The workhorse model in this literature is the hedonic price model (HPM). Following Rosen (1974) and Harrison and Rubinfeld (1978), the HPM assumes that consumers maximize utility over a composite good and a bundle of housing characteristics. In equilibrium, the market price of a dwelling reflects the implicit prices of its constituent attributes: structural features (number of rooms, lot size), locational characteristics (distance to employment centers, school quality), and neighborhood amenities (parks, crime rates, environmental quality). The selling price is thus a composite of all these attributes, and regression analysis can recover their individual implicit values.

The hedonic model's theoretical foundation is well understood. A housing unit represents several consumption choices at once: a location decision that defines access to education systems, employment markets, and environmental externalities, as well as a piece of architecture with size, quality, and maintenance characteristics. As notes in a selective review, HPMs have been applied to value everything from school quality to environmental goods. The key challenge is functional form: the relationship between housing attributes and prices is generally nonlinear and may involve complex interactions.

Recent work has complemented traditional HPMs with machine learning techniques—such as random forests, gradient boosting, and neural networks—to capture nonlinear and interaction effects. argue that an objective approach to quantifying microscopic-level perception from pixel information in streetscape images can complement people's subjective sensory processes regarding the environment. This represents a new frontier in which big data and deep learning enhance the traditional hedonic approach.

A key point for the purposes of this review is that the hedonic model provides the microfoundation for quality-adjusting housing: if one can estimate the implicit prices of all dwelling and neighborhood attributes, one can in principle convert any heterogeneous dwelling into an equivalent quantity of standardized "housing services." We return to this point in Part II.

1.2 Transportation Access and Commute Quality

One of the earliest and most robust findings in urban economics is that proximity to employment and transportation infrastructure is capitalized into property values. The monocentric city model (Alonso, 1964; Mills, 1967; Muth, 1969) predicts that housing prices decline with distance from the central business district (CBD), reflecting commuting cost trade-offs.

Heikkila et al. (1989) used a hedonic model with multiple employment centers to examine land values in Los Angeles, and Waddell et al. (1993) added distances to other regional amenities to predict housing prices. Both found distances to employment subcenters to be significant predictors, but arrived at the provocative conclusion that distance to the CBD may no longer be a significant determinant of land values in polycentric metropolitan areas. Katz and Rosen (1998) found that commuting *time* to the San Francisco CBD was significant even when raw distance was not, highlighting the importance of measuring effective accessibility rather than simple Euclidean distance.

More recently, accessibility indices—combining information on travel times, road networks, and employment distributions—have been shown to substantially improve hedonic model performance. A study on Madrid found that optimal car and walk accessibility indices improved regression accuracy by 13% and random forest accuracy by 21.6%, while reducing spatial autocorrelation by roughly 35% (see e.g., ScienceDirect, 2023).

1.2.1 Rail Transit and Property Values

A large sub-literature examines the capitalization of proximity to rail transit stations. studied Portland, Oregon, finding positive capitalization of proximity to light-rail transit (LRT) stations for homes within 500 meters of walking distance. A systematic review of studies from 2016–2024 finds that rail systems generally confer a 4–20% price premium to nearby properties, with effects peaking within 500 meters and varying by transit type (heavy rail > light rail), urban density, and local amenities.

Importantly, the transit–price relationship is not uniformly positive. Acuña (2023) studied the staggered opening of new Metro stations in Prince George's County, a suburb of Washington, DC, and found that housing prices actually *increased* with distance from stations. The preferred repeat-sales model estimated a marginal price increase of 4.6% for a one-mile increase in distance, suggesting that in suburban, automobile-dependent settings, proximity to rail is primarily viewed as a disamenity. This underscores that the value of transit access is highly context-dependent.

1.3 Street Network Connectivity and Intersection Density

Street network form—the degree to which streets are interconnected, the density of intersections, and block size—has attracted growing attention as a determinant of both travel behavior and property values. American street network intersection densities typically range from about 60 intersections per square mile (e.g., downtown Salt Lake City) to over 400 in dense, traditional urban cores. LEED for Neighborhood Development (LEED-ND) uses 140 intersections per square mile as a standard for walkable design.

Intersection density serves as a proxy for walkability, pedestrian convenience, and route choice. Smaller blocks and more intersections create more direct pedestrian paths, support more frequent transit stops, and generate a more human-scaled environment. A key theoretical contribution comes from Barrington-Leigh and Millard-Ball (2015, 2020), who conceptualize density as a property of street network connectivity: neighborhoods with poor road connectivity are unlikely to experience densification over time due to the path-dependence of street infrastructure.

The relationship between street connectivity and property values is nuanced. The Victoria Transport Policy Institute notes that residential properties tend to have lower values on connected streets than on cul-de-sacs, reflecting privacy and traffic-calming preferences. However, this may be offset where New Urbanist design practices incorporate appropriate traffic control and pedestrian-friendly features. Song and Knaap (2003), Li and Brown (1980), and Matthews and Turnbull (2007) have all explored individual walkability components and their effects on housing prices, with mixed but generally positive findings for well-designed connectivity.

provides an economic perspective on street width, estimating that in the urbanized portions of 20 large US counties, residential street rights-of-way average 55 feet wide—far greater than the functional minimum of 16 feet. The land value of residential streets totals $959 billion in his sample. In high-cost markets like Santa Clara County, narrowing rights-of-way to 16 feet could save over $100,000 per housing unit through reduced land consumption.

1.4 Local Amenities: Parks, Schools, Safety, and Environmental Quality

A rich hedonic literature has documented significant price effects for a range of neighborhood amenities. In Columbus, Ohio, physical disorder (graffiti, trash, unkempt lawns, abandoned buildings, busy streets) was capitalized into home prices, while in Baltimore, neighborhood racial composition and socioeconomic status were more strongly related to housing values than crime or school quality. This heterogeneity across metropolitan areas is a consistent finding.

School quality is among the most reliably capitalized amenities. Black (1999) pioneered regression discontinuity designs to isolate the effect of school attendance boundaries on housing prices, finding substantial premiums for access to higher-performing schools. Proximity to parks and open space also commands premiums: a study in Southold, Long Island found that properties adjacent to open space had, on average, 12.8% higher per-acre values.

Environmental quality broadly defined—air quality, noise, water proximity, and greenness—has been extensively studied. valued urban greenness in Lisbon using remote sensing and hedonic methods. A growing number of studies use the Normalized Difference Vegetation Index (NDVI) to measure greenness and find generally positive associations with property values.

1.5 Streetscape Perception and Visual Quality

A newer strand of literature uses Google Street View (GSV) images and deep learning to measure the visual quality of streets from the pedestrian perspective. This micro-scale approach captures attributes that traditional top-down variables (e.g., the green ratio measured from satellite imagery) miss.

Studies in Beijing and Shanghai using this methodology found that street-level greenness (the Green View Index, GVI) and sky openness (the Sky View Index, SVI) both positively affect housing prices. The marginal implicit prices were substantial—RMB 39,377 for greenness in Beijing and RMB 6,011 in Shanghai. Some studies report negative associations between certain streetscape elements and prices, suggesting that the relationship depends on context and measurement approach. Work on Korean metropolitan areas using deep learning and street view imagery found that streetscape effects on single-family home prices are highly nonlinear and involve significant interaction effects, arguing that interpretable machine learning methods (e.g., SHAP values from gradient boosting models) are essential for understanding these complex relationships.

1.6 Walkability and Composite Indices

Walk Score and similar composite walkability indices have been widely adopted. Pivo and Fisher (2011), Rauterkus and Miller (2011), and Li et al. (2015) have found positive associations between composite walkability and property values. A Melbourne, Australia study found that walkability was positively associated with house prices across all neighborhood socioeconomic quintiles, with destination accessibility driving higher prices and poor transit access reducing them.

synthesized the literature on pedestrian- and transit-oriented development, finding consistent price premiums for walkable, mixed-use neighborhoods. The H+T Index (Housing + Transportation) developed by the Center for Neighborhood Technology captures the joint cost of housing and transportation, revealing that housing costs are higher but travel costs are lower in denser, more multimodal urban areas—a trade-off often invisible when housing affordability is assessed in isolation.

1.7 The 'Neighborhood Brand' Effect

An emerging strand treats the neighborhood itself as a heuristic that aggregates many local attributes into a single "brand." A 2024 study using over 50,000 housing transactions in Charleston, South Carolina estimated neighborhood fixed effects within a hedonic framework. The neighborhood fixed effect captures time-invariant local quality—sidewalks, natural amenities, water quality, transport access, homeowner associations, community centers—and was found to exert a pronounced effect on regression results, outperforming alternative areal units such as census tracts. The fixed-effect estimates effectively captured the wide range of house price premiums, from high-priced historic districts to low-price, poverty-stricken areas.

1.8 Market Access and Housing Prices

A theoretically grounded alternative to ad hoc accessibility measures is the concept of *market access*, derived from general equilibrium trade theory. Rather than measuring proximity to a single center or a set of amenities, market access aggregates a location's access to *all* other locations, weighted by their economic mass and discounted by bilateral trade or commuting costs. The concept originates in the Eaton–Kortum (2002) trade framework and was brought to bear on spatial economics by Donaldson and Hornbeck (2016) and applied within cities by Ahlfeldt et al. (2015) and Heblich et al. (2020).

1.8.1 The Donaldson–Hornbeck Framework

Donaldson and Hornbeck (2016), published in the *Quarterly Journal of Economics*, develop the market access approach to measure the aggregate impact of railroads on the 19th-century US economy. They derive from an Eaton–Kortum model (extended to allow for labor mobility) a reduced-form expression in which equilibrium land rents in county o are log-linear in a single endogenous variable: market access MA_o. Formally:

$$MA_o = \sum_d \tau_{od}^{-\theta} \cdot Y_d \tag{1}$$

where τ_{od} is the bilateral trade cost between origin o and destination d, θ is the trade elasticity, and Y_d captures the economic mass (income or population) of destination d. This measure captures both firms' desire to sell goods elsewhere at a high price and consumers' desire to buy goods from elsewhere at a low price.

The key theoretical result is that *all* general equilibrium effects of changes in transportation infrastructure on a location's economy—including indirect effects operating through changes in other locations—are summarized by changes in that location's market access. Using a network database of railroads and waterways and Dijkstra's algorithm to compute lowest-cost county-to-county freight routes, Donaldson and Hornbeck estimate that changes in county market access were capitalized into agricultural land values with an elasticity of approximately 1.1 as the railroad network expanded from 1870 to 1890. Removing all railroads in 1890 would have decreased the total value of US agricultural land by 64%.

While the original application was to agricultural land, the theoretical framework applies equally to any immobile factor whose returns depend on access to markets. For residential housing, the analog is straightforward: a location's residential land value should be increasing in its "residential market access"—its access to employment opportunities (for commuters), consumption amenities (for residents), and the broader urban economy.

1.8.2 Within-City Market Access: Ahlfeldt et al. (2015)

Ahlfeldt, Redding, Sturm, and Wolf (2015), published in *Econometrica* and awarded the Frisch Medal, bring the market access concept inside the city. They develop a quantitative model of internal city structure with agglomeration and dispersion forces across thousands of heterogeneous city blocks in Berlin. Idiosyncratic shocks to commuting decisions yield a gravity equation for commuting flows, providing microfoundations for the link between market access and floor-space prices.

Using the exogenous variation from Berlin's division (1961) and reunification (1989), they show that blocks in West Berlin closer to the pre-war concentration of economic activity in East Berlin experienced larger declines in land prices during division and larger recoveries after reunification. The estimated elasticity of floor-space prices with respect to changes in effective market access—operating through both production externalities (access to employment density) and residential externalities (access to population density)—is substantial. They estimate an elasticity of productivity with respect to workplace employment density of 0.07 and an elasticity of amenities with respect to residential density of 0.14, both highly localized and decaying rapidly with travel time.

The Ahlfeldt et al. framework makes explicit that what drives residential property values is not distance to a single center but a location's *commuter market access*—the sum of wage-weighted employment opportunities accessible via the commuting network, discounted by commuting costs. This provides a structural interpretation for the reduced-form hedonic findings on transit access and commute quality reviewed in Section 1.2: proximity to rail stations or highway ramps matters because these infrastructure elements increase a location's market access.

1.8.3 Further Applications

Heblich, Redding, and Sturm (2020) apply the quantitative spatial framework to London, using the construction of the underground railway in the 19th century as a source of variation. They show that the railway dramatically increased market access for locations near stations, leading to substantial increases in both population density and property values. The model implies that commuter market access is the key channel: locations that gained better access to the rest of London's labor market saw the largest increases in residential land values.

Ahlfeldt (2013) applies a gravity-type labor-market accessibility model to Greater London to investigate house price capitalization effects, finding that the spatial scope of labor-market accessibility effects is substantial and that accessibility measures grounded in economic geography outperform simple distance-to-CBD measures in hedonic models.

More broadly, the quantitative spatial economics literature surveyed by Redding and Rossi-Hansberg (2017) and Redding (2024) shows that market access—whether defined for goods trade (as in Donaldson–Hornbeck) or for commuting (as in Ahlfeldt et al.)—is a sufficient statistic for the general equilibrium value of a location's connectivity to the rest of the spatial economy. For housing prices specifically, this means that a location's market access captures the aggregate effect of all the individual accessibility dimensions (transit proximity, street connectivity, commute time, employment access) studied in the hedonic literature—but does so in a theoretically disciplined way that accounts for general equilibrium feedbacks.

The market access framework also connects naturally to the effective housing supply concept developed in Part II. A dwelling's "effective" contribution to the housing stock depends not only on its physical attributes but on its location's market access: the same structure provides more housing services (broadly defined) when it is situated in a high-market-access location. Quality-adjusting housing supply using market access weights would capture this locational component of housing value.

2 Housing Quality as a Component of Effective Housing Supply

2.1 Motivation: The Analogy to Effective Labor Supply

In labor economics, heterogeneous workers are routinely aggregated into "effective labor supply" using relative wages as weights. A worker earning twice the average wage contributes two "efficiency units" of labor. This allows models to treat labor as a scalar input even when the underlying workforce is heterogeneous in skill, education, and productivity. Formally, effective labor is:

$$L^{\text{eff}} = \sum_i \frac{w_i}{\bar{w}} \cdot L_i \tag{2}$$

where w_i is the wage of worker type i, $\bar{w}$ is the average wage, and L_i is the count of type-i workers.

An analogous question arises for housing: dwelling units are enormously heterogeneous—in size, structural quality, age, amenities, and neighborhood characteristics. Can they be aggregated into "effective housing supply" units using hedonic price weights? Formally, one might define:

$$H^{\text{eff}} = \sum_j \frac{\hat{p}_j}{\bar{p}} \cdot 1 \tag{3}$$

where $\hat{p}_j$ is the hedonic-predicted value of dwelling j (reflecting all its quality attributes) and $\bar{p}$ is the average value, so that each dwelling contributes its quality-adjusted share to aggregate housing services. Equivalently, if P denotes the per-unit price of housing services in a market, then $h_j = \hat{p}_j/P$ is the quantity of "housing services" supplied by dwelling j, and $H^{\text{eff}} = \sum_j h_j$.

This section reviews the theoretical foundations and empirical progress on this concept, connecting the hedonic literature on neighborhood quality (Section 1) to the spatial equilibrium literature on housing supply and equilibrium prices.

2.2 The "Housing Services" Abstraction in Spatial Equilibrium Models

Most spatial equilibrium models—including the foundational Rosen (1979)–Roback (1982) framework and its extensions by Moretti (2011), Hsieh and Moretti (2019), and others—treat housing as a scalar. Each worker consumes some quantity of "housing services" h at a per-unit price P_i in city i. The heterogeneity of actual dwelling units is collapsed into this single dimension.

In the Rosen–Roback framework, spatial equilibrium requires that mobile workers are indifferent across locations: higher wages in productive cities are offset by higher housing costs and/or lower amenities. Formally, utility equalization requires:

$$V = \frac{W_i \cdot Z_i}{P_i^{\beta}} \quad \text{for all cities } i \tag{4}$$

where W_i is the wage, Z_i captures amenities, P_i is the housing price, and β is the expenditure share on housing. This condition, combined with labor demand and housing supply equations, jointly determines area population, wages, and prices as functions of exogenous differences in productivity, amenities, and construction costs (Glaeser and Gottlieb, 2009).

The critical point is that P_i in this framework is implicitly a *quality-adjusted* price: the price per unit of housing services. The model assumes that all quality variation has been absorbed so that P_i is comparable across locations. In practice, this is a strong assumption. Two cities with the same P_i might have very different housing stocks—one with large, new units in walkable neighborhoods with good schools, the other with smaller, older units in car-dependent areas. The "effective housing supply" question asks: given that quality varies, how should we aggregate the stock?

2.3 Early Hedonic Approaches to Measuring Housing Quantity

The earliest formal attempt to use hedonic regressions to construct quality-adjusted housing quantities is Barnett (1979), produced for the RAND Housing Assistance Supply Experiment (HASE). Barnett's approach treats the hedonic regression as providing the "exchange rates" between different housing attributes:

> "Weighting the attributes of dwellings by [hedonic] prices, disparate attributes such as the number of rooms, type of heating system, and quality of the neighborhood can be summed to yield measures of services supplied by dwellings that are comparable across dwellings and over time."

This is precisely the housing analog to computing effective labor in efficiency units. By estimating a hedonic price function from Brown County, Wisconsin rental data and using the coefficients as weights, Barnett constructed a "housing services" index for each dwelling. The sum across all dwellings in a market yields an aggregate quantity of effective housing supply.

The same logic underlies the Bureau of Labor Statistics' hedonic quality adjustment for the CPI shelter component. The BLS estimates hedonic regression models with structural characteristics, neighborhood variables, services included in rent, and dwelling age to separate "pure" price changes from quality changes. The quality component is precisely what the "effective housing supply" concept captures.

2.4 The Baum-Snow and Han (2024) Decomposition

The most explicit recent decomposition of housing supply responses into units versus quality is Baum-Snow and Han (2024), published in the *Journal of Political Economy*. They perform a comprehensive neighborhood-level analysis of housing supply across 306 U.S. metro areas and decompose supply responses of total housing services into:

1. **Units:** the number of dwelling units, further decomposed into new construction on previously undeveloped land versus redevelopment (teardowns and rebuilds) of existing developed areas.
2. **Quality:** changes in floor space per unit, renovations, and upgrades to existing structures.

They measure total housing services using floor space as a physical proxy for the quality-weighted aggregate. Their key finding is that predicted floor space supply elasticities average 0.5 across all urban neighborhoods in the United States, while housing *unit* supply elasticities average only 0.3. The gap—floor space elasticity exceeding unit elasticity—indicates that quality upgrading (bigger, better units) is an important margin of supply response beyond simply adding units. New construction accounts for about 50% of unit supply responses, with important additional roles for teardowns and renovations.

Crucially, these elasticities exhibit greater variation *within* metro regions than *between* them. Supply responses grow with CBD distance, mostly due to the increasing availability of undeveloped land, flatter terrain, and less stringent regulation. This microgeographic heterogeneity means that aggregate metro-level supply elasticities (of the kind estimated by) are an average over very different neighborhood-level responses.

An important insight for the "effective housing supply" concept is that aggregation from micro to macro elasticities incorporates substitution patterns across neighborhoods. As neighborhoods become stronger demand substitutes, a shock affecting one location disperses housing demand to a wider range of areas, opening up supply-elastic neighborhoods. The result is macro supply elasticities that exceed typical tract-level elasticities.

2.5 "Home Productivity" as a Supply-Side Quality Concept

introduce the concept of "home productivity" into the spatial equilibrium framework. In their neoclassical model of location, each city j is characterized by three exogenous attributes: trade productivity A_j^X (determining wages), quality of life Q_j (amenities), and home productivity A_j^Y—the efficiency with which land and capital are converted into housing services.

Higher home productivity means that a given quantity of land and construction capital produces more housing services, lowering housing costs for a given demand level. This is the supply-side analog to what the hedonic literature measures on the demand side: a city with high home productivity provides more "effective housing" per unit of physical stock. Albouy and Stuart estimate home productivity differences across U.S. metros using a cross-section of wages, rents, population, and land area, finding substantial variation.

The concept of home productivity connects directly to the neighborhood quality literature in Section 1. Better street connectivity, transit access, walkability, school quality, and environmental amenities all increase the "effective" housing services that a given dwelling provides—not by changing the physical structure, but by enhancing its locational and neighborhood attributes. From the worker's perspective, a dwelling in a walkable, transit-accessible neighborhood with good schools provides more utility per dollar of rent than an identical structure in a car-dependent area with poor amenities. In this sense, neighborhood quality is a form of "home productivity."

2.6 Housing Supply Constraints and Spatial Misallocation

The most influential paper connecting housing supply constraints to aggregate economic outcomes is Hsieh and Moretti (2019). They argued that high-productivity cities like New York and the San Francisco Bay Area have adopted stringent restrictions on new housing supply, effectively limiting the number of workers who can access the high productivity available there. Using a spatial equilibrium model and data from 220 metropolitan areas, they estimated that these constraints lowered aggregate US growth by 36% from 1964 to 2009.

The mechanism operates through the "local price" $Q_i \equiv P_i^\beta / Z_i$—the ratio of housing costs to amenities. In equilibrium, the nominal wage is proportional to this local price. When housing constraints prevent workers from moving to high-productivity cities, the dispersion of nominal wages across space widens, indicating growing misallocation. Aggregate output is a power mean of local TFP weighted by the inverse of local prices: a mean-preserving spread in local prices lowers aggregate output.

For the effective housing supply concept, the Hsieh–Moretti framework highlights an important point: what matters for labor allocation is not just the *quantity* of housing units but their *cost per unit of service*—i.e., the effective price. A city that provides the same number of units but at lower quality (worse neighborhoods, longer commutes, poorer schools) effectively has a higher cost of living per unit of utility, deterring in-migration just as higher rents would.

It is important to note that the Hsieh–Moretti estimates have been subject to scrutiny. Caplan (2021) identified arithmetical inconsistencies. More recently, Greaney (2023) found coding errors and a scale-dependence problem: when corrected, the headline result drops dramatically, with liberalizing three superstar cities boosting national GDP by only about 0.02%. The debate remains active, but the conceptual point—that housing constraints impede efficient spatial allocation of labor—is broadly accepted.

2.7 Housing Supply Elasticity and the Quality Margin

A central finding in the housing supply literature is that the elasticity of housing supply varies enormously across locations and is a key determinant of how productivity shocks translate into prices versus quantities. provided canonical metro-level estimates, showing that geographic constraints (steep terrain, bodies of water) and regulatory restrictions jointly determine local supply responsiveness. Glaeser, Gyourko, and Saks (2006) documented that land-use regulations, not construction costs or land scarcity per se, explain the bulk of cross-metro differences in housing supply.

What the effective housing supply concept adds is that the *quality* margin is a distinct and important component of supply adjustment. When demand increases in a neighborhood, the response may take several forms:

- New construction on undeveloped land (extensive margin of units);
- Teardowns and rebuilding at higher density or quality (intensive margin of units);
- Renovation and upgrading of existing structures (quality margin);
- Neighborhood investment that improves amenities, schools, transit, walkability (locational quality margin).

The first two are captured by unit supply elasticities; the latter two are captured only by a quality-adjusted or "effective" supply elasticity. As show, the quality margin accounts for a significant fraction of total supply response: their floor-space elasticity (0.5) exceeds their unit elasticity (0.3) by about two-thirds. The implication is that models treating housing supply as a simple count of units may substantially understate the true responsiveness of the housing market to demand shocks.

Moreover, the durable and immobile nature of housing stock creates asymmetry. show that houses are built more quickly than they depreciate, making supply relatively inelastic in declining areas: prices fall but the stock persists. This "ratchet effect" means that quality can deteriorate (through deferred maintenance, neighborhood decline) even as units remain in the stock—driving a wedge between the count of units and the effective supply of housing services.

2.8 Amenity Capitalization, Worker Sorting, and Effective Supply

The hedonic evidence from Section 1 feeds directly into the effective housing supply concept. The capitalization of neighborhood quality into housing prices is the empirical manifestation of quality-adjustment: a dwelling in a walkable neighborhood with good schools and transit access commands a higher price precisely because it provides more housing services (broadly defined) per unit.

Diamond (2016) built an equilibrium model showing that high-skilled workers increasingly concentrate in high-wage, high-cost, high-amenity cities, while lower-skilled workers are priced out. Albouy and Faberman (2025) introduce life-cycle dynamics, finding that workers move toward lower-amenity (and lower-cost) areas during their thirties and forties, driven by having children. Since local amenities are capitalized into housing prices, individuals move to lower-cost locations to avoid paying for amenities they are not consuming. This is precisely the logic of effective housing supply: the same physical unit provides different effective services to different household types at different life stages.

provide an authoritative survey of equilibrium sorting models in the *Journal of Economic Literature*. They show how households sort across neighborhoods according to wealth and preferences for public goods, social characteristics, and commuting opportunities. The sorting process endogenously influences amenity supply. Their framework can predict how equilibrium prices and quantities respond to policies targeting product quality or amenity provision—precisely the kind of analysis enabled by an effective housing supply concept.

2.9 The Gap in the Literature and Research Opportunities

Despite the intellectual ingredients being in place, there is **no widely adopted, formalized "effective housing supply" index** that is as standard as efficiency-weighted labor supply in labor economics. The concept exists in pieces:

1. **Hedonic quality adjustment** (; BLS CPI methods; IMF RPPI Handbook) provides the "exchange rates" for converting heterogeneous attributes into comparable units.
2. **Floor space as a physical proxy** () captures one dimension of quality but not the full bundle of neighborhood attributes.
3. **"Home productivity"** () provides a supply-side efficiency parameter in spatial equilibrium models, but is estimated as a residual rather than constructed from observable quality measures.

4. **Constant-quality price indices** (BLS, Eurostat, IMF) implicitly separate price from quality changes, but the quality component is used for index construction rather than for measuring effective supply.

Several factors explain why the analogy is harder for housing than for labor:

Durability and immobility. Existing housing stock depreciates slowly, and you cannot simply "add quality" the way a worker can acquire skills. The stock at any moment reflects decades of construction history. When the housing depreciation rate is low, stocks adjust slowly to changes in demand and supply conditions, slowing down the adjustment processes that spatial equilibrium models rely on (Glaeser and Gyourko, 2005).

Multidimensionality. A dwelling's quality is multidimensional in a way that is harder to collapse than labor productivity. Structural quality, neighborhood quality, access quality (commute), environmental quality, and school quality may be valued very differently by different household types. Albouy et al. (2016) develop a demand system estimating that the price elasticity of housing demand is about 2/3 and the expenditure elasticity is also about 2/3, implying that housing expenditure shares *rise* with house prices—complicating simple aggregation.

Endogeneity of quality. Neighborhood quality is partly endogenous to the sorting process itself. As high-income, high-skill workers concentrate in a neighborhood, amenities improve endogenously (better restaurants, retail, schools funded by property taxes), further increasing the effective housing supply in that location. This feedback loop between sorting and amenity provision makes quality-adjusted measurement more complex than in labor markets.

A research opportunity. The hedonic valuations of street quality, connectivity, transit access, school quality, greenness, and other amenities documented in Section 1 could, in principle, be used to construct quality-adjusted housing supply measures that would enter spatial equilibrium models with richer implications for labor allocation and equilibrium prices. Combining the Baum-Snow and Han (2024) decomposition with the streetscape/neighborhood quality literature would allow researchers to assess how much of the "effective housing supply" response to demand shocks comes from physical construction versus locational quality improvements—and how this decomposition varies across neighborhoods, cities, and over time.

3 Conclusion

The two bodies of literature reviewed here are intimately connected but rarely integrated. The hedonic price literature (Section 1) provides rich micro-level evidence on how neighborhood and street characteristics—transportation access, street connectivity, amenities, streetscape quality, walkability—are capitalized into property values. These implicit prices are precisely the "weights" needed to construct quality-adjusted housing supply measures.

The spatial equilibrium literature (Section 2) shows that housing supply—its quantity, quality, and elasticity—is a critical determinant of equilibrium wages, labor allocation, and aggregate productivity. Most models in this tradition treat housing as a scalar, abstracting from quality heterogeneity. The concept of "effective housing supply"—aggregating heterogeneous dwellings into efficiency units using hedonic price weights, analogous to effective labor supply—exists in fragmentary form but has not been formalized as a standard tool.

Bridging this gap represents a productive research frontier. An effective housing supply index, built from the hedonic micro-evidence and embedded in spatial equilibrium models, would allow researchers and policymakers to assess not only how many dwellings are supplied, but how much *housing service*—broadly inclusive of neighborhood quality—the stock actually provides. This has direct implications for understanding the welfare effects of land-use regulation, the spatial allocation of labor, and the distributional consequences of neighborhood change.

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